

Challenge (Habanero)

Q1. Simplify $\sqrt{64}$

- (A) 8
- (B) 4
- (C) 16
- (D) 32
- (E) 6

Q2. If $x = \sqrt[3]{8}$ and $y = \sqrt{16}$, what is $x + y$?

- (A) 6
- (B) 4
- (C) 8
- (D) 10
- (E) 12

Q3. A hiker travels $\sqrt{25}$ miles to reach the base of a mountain. How many miles did the hiker travel?

- (A) 5
- (B) 10
- (C) 15
- (D) 20
- (E) 25

Q4. What is the simplified form of $\sqrt{75}$?

- (A) $5\sqrt{3}$
- (B) $3\sqrt{5}$
- (C) $25\sqrt{3}$
- (D) $15\sqrt{2}$
- (E) $10\sqrt{5}$

Q5. A car's fuel consumption is $\sqrt{36}$ gallons per hour. How

many gallons of fuel does it consume per hour?

- (A) 6
- (B) 8
- (C) 10
- (D) 12
- (E) 15

Q6. A population of deer grows at a rate of $\sqrt[3]{27}$ per year. What is the growth rate?

- (A) 3
- (B) 6
- (C) 9
- (D) 12
- (E) 15

Q7. What is the value of x in $x = \sqrt{121}$?

- (A) 10
- (B) 11
- (C) 12
- (D) 13
- (E) 14

Q8. A park ranger measures the length of a trail to be $\sqrt{49}$ meters. What is the length of the trail?

- (A) 5
- (B) 6
- (C) 7
- (D) 8
- (E) 9

Open Ended Questions (FRQ)

Q9. Simplify the expression $\sqrt{48} + \sqrt{18}$. Show all work and simplify your answer.

Q10. A group of friends want to go on a hike that is $\sqrt{100}$ miles long. If they have already hiked $\sqrt{25}$ miles, how many more miles do they have to hike? Show all work and explain your answer.

Chef's Tips

- Make sure students show all work for FRQs
- Encourage students to use radicals to simplify expressions
- Remind students to check their units in word problems